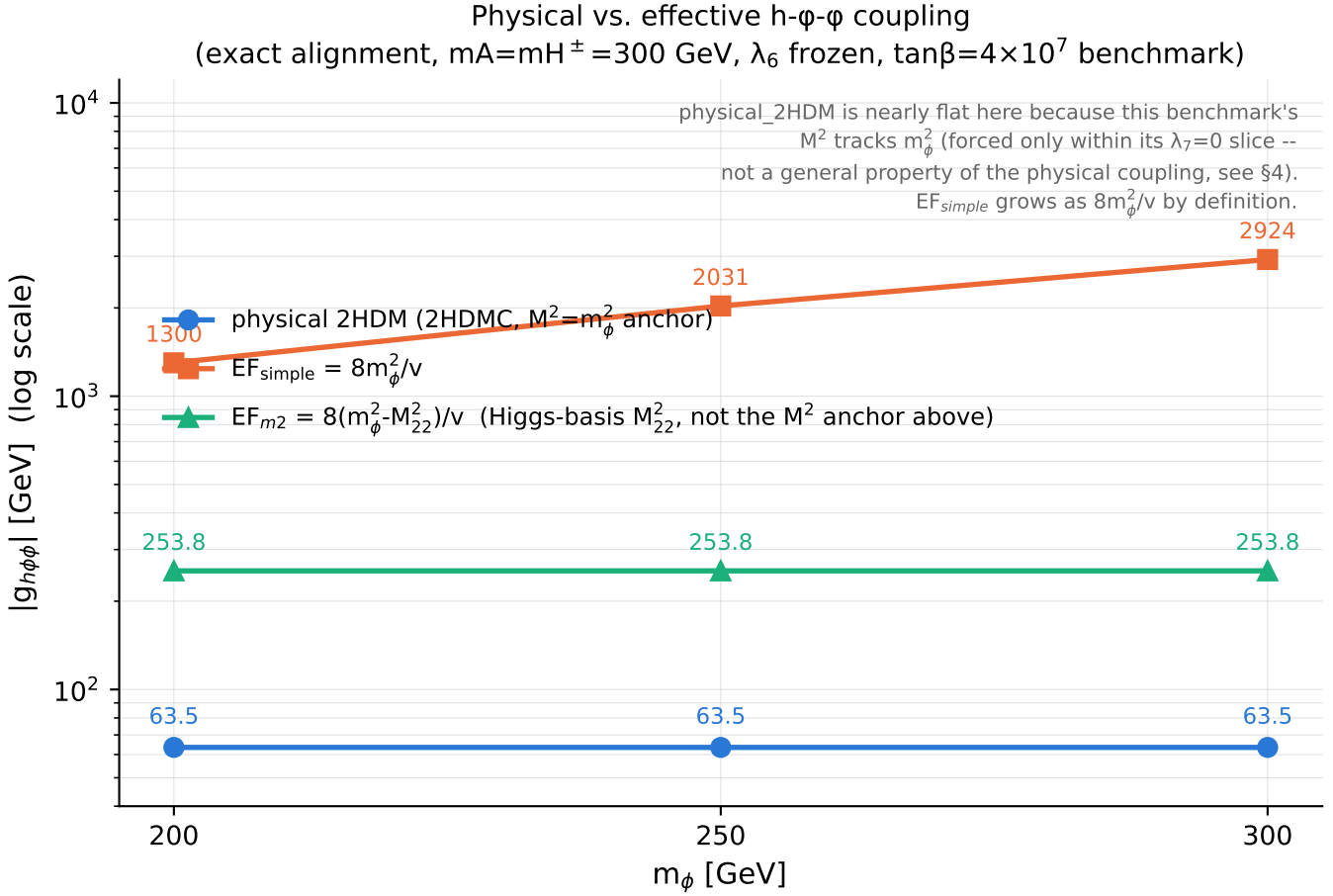
Physical vs. effective h-φ-φ coupling
(exact alignment, mA=mH$^{\pm}$=300 GeV, λ_6 frozen, tanβ=4×10^7 benchmark)
physical_2HDM is nearly flat here because this benchmark's
M^2 tracks m_ϕ^2 (forced only within its λ_7=0 slice --
not a general property of the physical coupling, see §4).
EF$_{simple}$ grows as $8m_\phi^2/v$ by definition.
physical 2HDM (2HDMC, M^2=m_ϕ^2 anchor)
EF$_{simple}$ = $8m_\phi^2/v$
EF$_{m2}$ = $8(m_\phi^2$-M$_{22}^2)/v$ (Higgs-basis M$_{22}^2$, not the M^2 anchor above)
1300
2031
2924
253.8
253.8
253.8
63.5
63.5
63.5
10^4
10^3
10^2
200
250
300
$|g_{h\phi\phi}|$ [GeV] (log scale)
m_ϕ [GeV]